Rietveld Refinement of a CVO Powder Diffraction Pattern

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***Abstract*—X-ray diffraction combined with Rietveld refinement provides a powerful method for determining crystallographic structures from samples. In this work, a Rietveld refinement of a calcium vanadate (CVO) sample was performed using FullProf. Starting from an initial structural model and experimental diffraction data, several structural and instrumental parameters were sequentially refined until convergence was achieved. The final refinement produced lattice parameters consistent with the orthorhombic structure and reasonable agreement factors. Additional data and figures related to the refinement process are available in the project repository.**

I. Introduction

X-ray diffraction is one of the most used techniques for determining the crystal structure of materials. When data is obtained from powdered samples, the measured pattern consists of overlapping reflections with random orientations. Extracting structural information from such patterns requires fitting techniques.

The Rietveld method is a refinement technique in which a calculated diffraction pattern derived from a crystallographic model is fitted to the experimental data by minimizing the difference between observed and calculated intensities [1]. During this process, structural, lattice, and instrumental profile parameters can be refined simultaneously.

Previous studies on CaV_2O_4 systems using Rietveld refinement have been reported in the literature [2]. In this work, the Rietveld method was applied to refine the crystal structure of a CVO sample using FullProf. The refinement procedure and the resulting structural parameters are presented and analyzed.

II. Methodology

The refinement was performed using FullProf. The experimental diffraction data was provided in DAT format, while the initial structural model was obtained from a CIF file containing the crystallographic parameters of the CVO structure. The diffraction pattern was measured in the approximate range of $15°$ to $100°$ in 2θ using Cu radiation, and the least squares minimization function was used.

To ensure stability, the refinement strategy followed a sequential approach as recommended in the literature [1]:

1) Refinement of the scale factor
2) Refinement of the background polynomial coefficients
3) Refinement of the zero-shift parameter
4) Refinement of lattice parameters
5) Refinement of peak-profile parameters (U, V, & W)
6) Refinement of atomic positions

The peak profiles were modeled using a pseudo-Voigt function, which is a linear combination of Gaussian and Lorentzian components commonly used to describe diffraction peak shapes in powder diffraction.

Each refinement cycle was executed using a maximum of 100 iterations. However, convergence was achieved in fewer iterations for all refinement stages.

III. Rietveld Refinement Results

Figure 1 shows the final refinement result. The red points correspond to the experimental diffraction data, the black line represents the calculated diffraction pattern obtained from the refined model, and the blue curve corresponds to the difference between observed and calculated intensities.

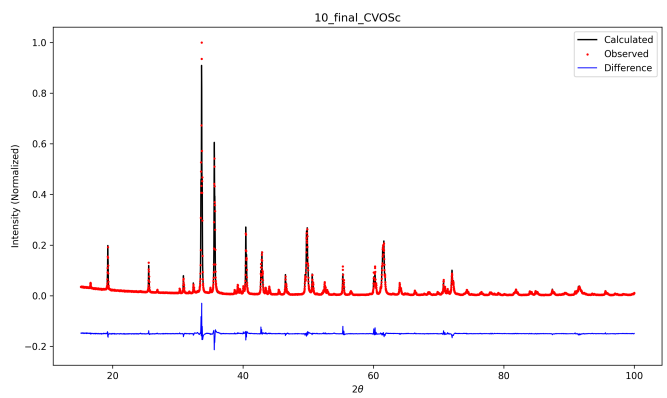

Fig. 1: Final Rietveld refinement showing observed data, calculated pattern, and difference curve.

The agreement between the calculated and experimental profiles indicates that the refined structural model reproduces the diffraction pattern with good accuracy. The difference curve oscillates around zero, indicating no major systematic deviations. The biggest difference is found on the main peak, indicating that further more specific refinement is needed to increase accuracy of the model.

IV. Peak Profile Analysis

To further evaluate the quality of the refinement, enlarged views of several representative diffraction peaks were analyzed. Figure 2 shows the four highest diffraction peaks.

The close agreement between the experimental points and the calculated profiles demonstrates that the pseudo-Voigt peak model accurately reproduces the peak shapes and positions.

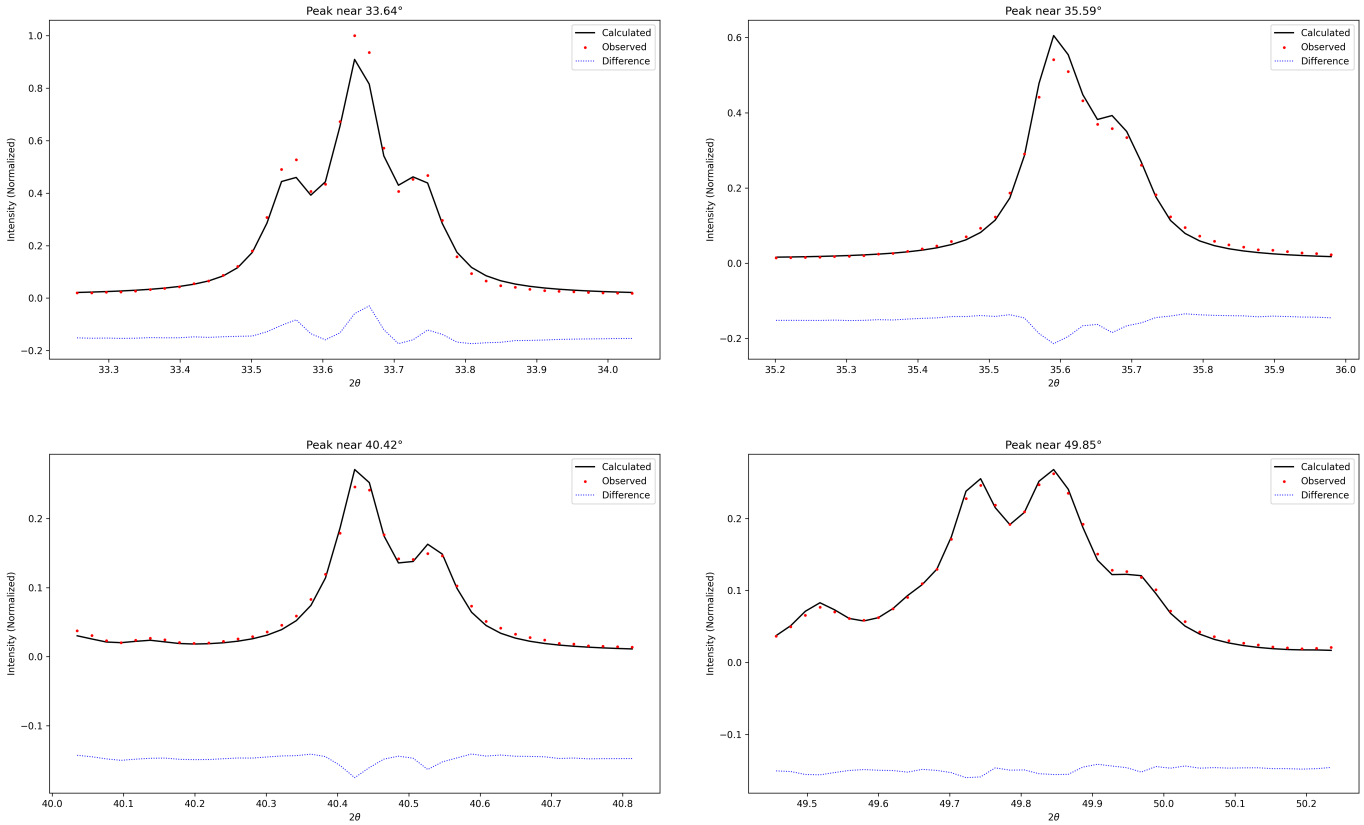

Fig. 2: Enlarged views of the four highest diffraction peaks showing the agreement between observed and calculated intensities.

V. Agreement Factors

The quality of the refinement is evaluated using several agreement factors.

The obtained values seen on the OUT file were:

- $R_p = 6.60\%$
- $R_{wp} = 8.69\%$
- $R_{exp} = 3.22\%$
- $\chi^2 = 7.30$

The parameter R_p represents the profile residual, which measures the difference between observed and calculated intensities. The weighted profile residual R_{wp} includes statistical weighting of each data point. The expected residual R_{exp} corresponds to the theoretical obtainable residual. Then, the overall success of the fitting is given by:

$$\chi^2 \sim \left(\frac{R_{wp}}{R_{exp}}\right)^2$$

This value gives a measure of the overall quality of the refinement. The final χ^2 value seen on the PCR file is 7.32. The small discrepancy between values on the OUT and PCR files can be attributed to rounding errors.

VI. Lattice Parameters

The refined lattice parameters obtained from the final refinement are:

- $a = 10.6754$ Å
- $b = 9.2087$ Å
- $c = 3.0086$ Å

The structure is orthorhombic with $\alpha = \beta = \gamma = 90°$. These parameters are consistent with the expected crystallographic structure of the material [2].

VII. Background Modeling

The background intensity was modeled using a sixth-degree polynomial. The refined coefficients were the following:

$$337.460,\ -528.651,\ 1116.252,$$

$$-1843.305,\ 1451.774,\ -404.450$$

These background coefficients account for contributions not related to Bragg reflections, such as instrumental noise.

VIII. Crystal Structure

The crystal structure obtained from the refined model is shown in Figure 3. The structure contains Ca (green), V (red), and O (blue) atoms arranged in an orthorhombic lattice. There are in total 8 V atoms, 4 Ca atoms and 16 O atoms inside a unit cell, making a total of 28 atoms per unit cell.

Each vanadium atom is bonded with 6 oxygen atoms (Figure 4), while each calcium atom is also bonded with 6 oxygen atoms (Figure 5)

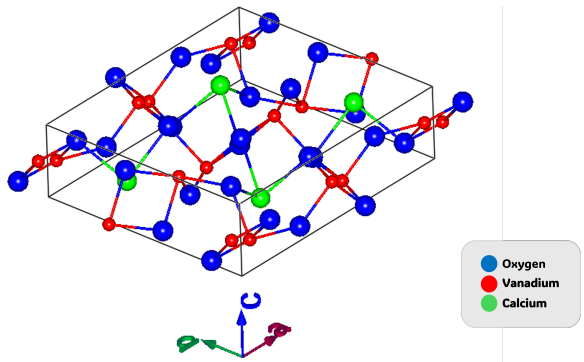

Fig. 3: Crystal structure showing bonds between nearest neighbours in a unit cell.

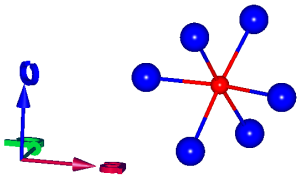

Fig. 4: Bonds between one V atom and 6 nearest O atoms.

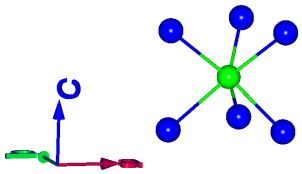

Fig. 5: Bonds between one Ca atom and 6 nearest O atoms.

IX. Repository and Supplementary Material

All files related to this project are available in the project repository:

Project GitHub repository

The repository contains the full refinement files, diffraction data, Python scripts used to generate the figures, and additional visualizations of the crystal structure from different perspectives. It also includes plots, data, and PCR files showing the evolution of the refinement process during intermediate stages.

X. Conclusion

The Rietveld refinement of the CVO powder diffraction pattern was successfully performed using FullProf. The refinement converged with reasonable agreement factors and produced reliable lattice parameters for the orthorhombic structure.

The refinement plot shown in figure 1 is a standard representation in Rietveld analysis [2].

The calculated diffraction pattern reproduces the experimental data with good agreement, confirming the validity of the refined structural model. These results demonstrate the effectiveness of the Rietveld method for extracting reliable structural information from experimental diffraction data.

References

[1] J. Rodríguez-Carvajal, *Introduction to the Program FullProf: Refinement of Crystal and Magnetic Structures*, Laboratoire Léon Brillouin (CEA-CNRS), France.

[2] S. R. Guitarra Bastidas, *Estudio estructural de compuestos tipo CaV_2O_4 mediante difracción de rayos X y refinamiento de Rietveld*, Master's thesis, Universidad San Francisco de Quito, Ecuador. Available at USFQ Digital Repository.